

MAT111 Take-home-mid

Deadline: 30th November, 2025, Time: 12:00 pm

Submit Hardcopy at my desk

Problem 1. Consider lists of length 6 made from the letters A, B, C, D, E, F, G, H . How many such lists are possible if repetition is not allowed and the list contains two consecutive vowels?

Problem 2. How many 7-digit binary strings begin in 1 or end in 1 or have exactly four 1's?

Problem 3. There is a very interesting function $\Gamma : [0, \infty) \rightarrow \mathbb{R}$ called the gamma function. It is defined as $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$. It has the remarkable property that if $x \in \mathbb{N}$, then $\Gamma(x) = (x-1)!$. Check that this is true for $x = 1, 2, 3, 4$. Notice that this function provides a way of extending factorials to numbers other than integers. Since $\Gamma(n) = (n-1)!$ for all $n \in \mathbb{N}$, we have the formula $n! = \Gamma(n+1)$. But Γ can be evaluated at any number in $[0, \infty)$, not just at integers, so we have a formula for $n!$ for any real number $n \in [0, \infty)$. Extra credit: Compute $\pi!$

Problem 4. Given a sphere S , a great circle of S is the intersection of S with a plane through its center. Every great circle divides S into two parts. A hemisphere is the union of the great circle and one of these two parts. Show that if five points are placed arbitrarily on S , then there is a hemisphere that contains four of them.

Problem 5. You are investigating a murder. The following facts have been established:

- (a) If Colonel Mustard is not guilty, then the crime took place in the library.
- (b) Either the weapon was the wrench or the crime took place in the billiard room.
- (c) If the crime took place at midnight, then Colonel Mustard is guilty.
- (d) Professor Plum is innocent if and only if the weapon was not the wrench.
- (e) Either Professor Plum or Colonel Mustard is guilty.

As a mathematician, determine what further evidence would conclusively determine the killer. For each of the following, explain how that piece of evidence — if established — would determine the killer.

- (a) The crime took place in the billiard room.
- (b) The crime did not take place in the billiard room.
- (c) The crime was committed at noon with a knife.
- (d) The crime took place at midnight in the kitchen.

Problem 6. At one of BracU MNS's parties, each man shook hands with everyone except his spouse, and no handshakes took place between women. If 13 married couples attended, how many handshakes were there among these 26 people?

Problem 7. A circular table has exactly 60 chairs around it. There are N people seated around this table in such a way that the next person to be seated must sit next to someone. What is smallest possible value of N ?

Problem 8. A mission to Mars will consist of 4 astronauts selected from 14 available. Exactly 5 of the 14 are trained in exobiology. If the mission requires at least 2 trained in exobiology, how many different crews can be selected

Problem 9. A spider has one sock and one shoe for each of its eight legs. In how many different orders can the spider put on its socks and shoes, assuming that, on each leg, the sock must be put on before the shoe?

Problem 10. Label one disk "1", two disks "2", three disks "3", . . . , fifty disks "50". Put these $1 + 2 + 3 + \dots + 50 = 1275$ labeled disks in a box. Disks are then drawn at random without replacement. What is the minimum number of disks that must be drawn to guarantee drawing at least ten disks with the same label.

Problem 11. Nian shoots free throws on a basketball court. He hits the first and misses the second, and thereafter the probability that he hits the next shot is equal to the proportion of shots he has hit so far. What is the probability he hits exactly 50 of his first 100 shots?

Problem 12. Assume that n is a positive integer. Prove that if one selects any $n + 1$ numbers from the set $\{1, 2, 3, \dots, 2n\}$, then two of the selected numbers will sum to $2n + 1$.

Problem 13. Assume that n is a positive integer. Prove that if one chooses any $n + 1$ distinct odd integers from $\{1, 2, 3, \dots, 3n\}$, then at least one of these numbers will divide another.

Problem 14. At a party, each person is acquainted with a certain number of others at the party (and is a stranger to everyone else). For example, Atonu may be acquainted with six people at the party while Masuda is acquainted with eight. Suppose that there are $n \geq 2$ people at a party. Prove that at least two people at this party have the same number of acquaintances at the party.

Problem 15. A magic square is an $n \times n$ matrix where the sum of the entries in each row, column and diagonal equal the same value. For example,

8	1	6
3	5	7
4	9	2

is a 3×3 matrix whose three rows, three columns, and two diagonals each sum to 15. Thus, this is a magic square. An antimagic square is an $n \times n$ matrix where each row, column and diagonal sums to a distinct value. For example,

1	2	3
8	9	4
7	6	5

is a 3×3 matrix whose rows sum to 6, 21 and 18, columns sum to 16, 17 and 12, and diagonals

sum to 21 and 15. Notice that all eight of these numbers is different that the rest, showing that this is an antimagic square. Prove that, for every n , there does not exist an $n \times n$ antimagic square where each entry is $-1, 0$ or 1 .

Problem 16. Towns $T_0, \dots, T_{n-1}$ are situated counterclockwise on a circular route of length 1. Assume that the town T_i can supply you with the amount of fuel that suffices to drive your car for distance p_i , and that $\sum_{i=0}^{n-1} p_i = 1$. That is, the total amount of fuel is exactly enough to take you once around the circle. Prove that you can start a car with an empty tank in one of the towns, drive counterclockwise around the entire circuit (picking up the fuel at each town on the route) and finish at the same spot (with an empty tank, of course).

Problem 17. The following conjectures are all false. Prove that they are false by finding a counterexample to each.

- (a) If x and y are real numbers, then $|x + y| = |x| + |y|$.
- (b) Suppose a, b and c are positive integers. If $a|bc$, then $a|b$ or $a|c$.
- (c) For every positive integer n ,

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} < 3$$

Problem 18. Suppose n is an integer. Prove that if $n^2|n$, then n is either $-1, 0$ or 1 .

Problem 19. Let $x, y > 0$. Prove that

$$\frac{x + y}{2} \geq \sqrt{xy}.$$

Problem 20. Suppose $\gcd(a, b) = 1$. Prove that for every integer m ,

$$\gcd(a, b + am) = 1.$$

Problem 21. Let m, n be integers with $\gcd(m, n) = d$. Prove that

$$\gcd\left(\frac{m}{d}, \frac{n}{d}\right) = 1,$$

and deduce that $m = dx_0$ and $n = dy_0$ for some coprime integers x_0 and y_0 .

Problem 22 (Frobenius Problem for two variables). Let a and b be coprime positive integers. Prove that the integer

$$ab - a - b$$

cannot be expressed as a nonnegative linear combination of a and b , but every integer greater than this value can.

Problem 23 (Wilson's Theorem). For every prime p , prove that $(p - 1)! \equiv -1 \pmod{p}$

Problem 24 (Fermat's Little Theorem). Prove that, for every prime number p , $a^p \equiv a \pmod{p}$

Problem 25. Let p be a prime. Show that if $x^k \equiv 1 \pmod{p}$ for all x , then $p - 1$ must divide k

Problem 26 (Chinese Remainder Theorem). Consider the following two modulo. $x \equiv 3 \pmod{11}$ and $x \equiv 5 \pmod{6}$. It is easy to find a solution, $x = 47$, by inspection. Here's another method. Since 6 and 11 are co prime, we can find a linear combination of 6 and 11 that equals one, for example, $(-1) \cdot 11 + 2 \cdot 6 = 1$. Now compute $5 \cdot (-1) \cdot 11 + 3 \cdot 2 \cdot 6 = -19$. This number is a solution modulo $6 \cdot 11 = 66$. Indeed, $47 \equiv -19 \pmod{66}$.

- a. Why does it work?
- b. Why the two moduli (Here, the two number 6, 11) need to be co prime? Can you give an example where they are not co prime and fails to give us a solution?
- c. Can you generalize this?
- d. Show that this algorithm can be extended to any finite number of simultaneous congruences, as long as the moduli are pairwise relatively prime.
- e. Show that there exist three consecutive numbers, each of which is divisible by the 1999th power of an integer.
- f. Show that there exist 1999 consecutive numbers, each of which is divisible by the cube of an integer.